A fixed 220-function dictionary for the local positive-angular C1 parent

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Use the source formulas and bounds of `fixed_box_c1.tex`. Thus $0 < x, \sigma \le 2^{-160}$, $0 < z \le 1$, all native gaps hold, $D, \beta \ge 0$, and $\lambda > 0$; $H = h/\sigma$ has no upper bound. The independent testing angles satisfy $|u_s|, |w_s| \le \delta = 2^{-100}$. This section completes the retained-square feature list for that rectangle. It does not cover $\lambda \le 0$ or other source/testing charts.

First apply the fixed testing gauge

$$(u, w, X, Y, Z, W) = (|u_s|, |w_s|, \operatorname{sgn}_+(w_s)X_s, \operatorname{sgn}_+(u_s)Y_s, \operatorname{sgn}_+(u_s)\operatorname{sgn}_+(w_s)Z_s, W_s),$$

where $\operatorname{sgn}_+(0) = 1$. Next multiply all four state coordinates by $\operatorname{sgn}_+(W)$, so $W \ge 0$. Both operations preserve the parent polynomial. Every testing function below is composed with this one fixed piecewise semialgebraic map. Put $Z_+ = \max(Z, 0)$ and $Z_- = \max(-Z, 0)$.

1 The exact whole-parent decomposition

All symbols $g, G, N_x, N_y, R, \Gamma, A_\theta, T_\theta, J_C, J_\ell$ have their explicit definitions in `fixed_box_c1.tex`. Set $F_A = u(Z - a_0 wW)$, $X_e = X + ewW$, and

$$a_C = cd_G/S, \quad q_C^{(0)} = \tfrac{1}{2} - a_C(2/\lambda + k), \quad p_C = q_C^{(0)} - a_C\delta.$$

The proved bounds give $0 \le a_C \le 6$ and $p_C > 1/4$. The circuit remainder satisfies

$$\mathscr{C} = \sum_{i=1}^{4} d_i^2 + \mathscr{C}_+,$$

and the angular difference is

$$\Delta_A = (2/\lambda + u + k)\mathscr{C} + u^2R^2 + \{2a_0u + (u+k)/L_0\}\Gamma^2 - A_\theta^2.$$

The complete parent is exactly

$$\begin{aligned} F_w = {} & g^{\mathsf{T}}Gg + \mathcal{E}_+ + (2+4m)w^2N_x^2 + \{4a + 2z(2C+1) - 1\}J_C^2 + (4\ell - 1)J_\ell^2 \\ & + \tfrac{1}{2}L_0F_A^2 + \frac{K_A + c - 1}{2}(A_\theta^2 + T_\theta^2) + \frac{K_A + c}{2}(A_\theta^2 - T_\theta^2) \\ & + \frac{cd_G}{S}\Delta_A + \{p_C + a_C(\delta - u)\}\mathscr{C}_+, \end{aligned}$$

where

$$\begin{aligned} \mathcal{E}_+ = {} & 4\beta(1+w^2)N_x^2 + 2(2D+1)(uX_e)^2 + 4bpzu^2\Gamma^2 \\ & + 4a(u^2 + w^2 + u^2w^2)J_C^2 + 4\ell u^2J_\ell^2 + 4Czw^2J_C^2. \end{aligned}$$

After the expansions below, every square coefficient outside G is nonnegative and source-only, except the explicitly separated testing factor $\delta - u$, which is nonnegative on the fixed testing rectangle. This equality is the full-state identity, not its restriction to a stationary state or a determinant. No inverse of D or β occurs.

2 Ordinary retained squares: seven additional functions

Let $\mathcal{V}$ be the twelve pre-Gram functions

$$\{X, Y, Z, W, uZ, wZ, uW, uwW, u^2W, u^{3/2}W, u^{1/2}W, \sqrt{uwWZ_+}\}.$$

All ordinary retained roots are source-linear combinations of $\mathcal{V}$ and the seven functions

$$wX,\ uwZ,\ uX,\ u^2wW,\ uY,\ wY,\ uwY.$$

Here are all seventeen roots; every displayed square root of a coefficient is source-only:

family	square roots
$\mathcal{E}_+$	$2\sqrt{\beta}N_x,\quad 2\sqrt{\beta}\,wN_x,\quad \sqrt{2(2D+1)}\,uX_e,\quad 2\sqrt{bpz}\,u\Gamma$
	$2\sqrt{a}\,uJ_C,\quad 2\sqrt{a}\,wJ_C,\quad 2\sqrt{a}\,uwJ_C,\quad 2\sqrt{\ell}\,uJ_\ell,\quad 2\sqrt{Cz}\,wJ_C$
N_x excess	$\sqrt{2+4m}\,wN_x$
radial excess	$\sqrt{4a+2z(2C+1)-1}\,J_C,\quad \sqrt{4\ell-1}\,J_\ell$
averaged square	$\sqrt{L_0/2}\,F_A$
angular excess	$\sqrt{(K_A+c-1)/2}\,A_\theta,\quad \sqrt{(K_A+c-1)/2}\,T_\theta$
angular difference	$\sqrt{(K_A+c)/2}\,qu(u-k)W,$
	$\sqrt{(K_A+c)/2}\,q\sqrt{2k}\sqrt{u}(u-k)W.$

For example $J_C = Y/q - bquW$ and $J_\ell = Y/q - wZ/q + QquW$, so multiplying by u, w, uw adds exactly the stated functions. All other rows follow by substitution. The last two squares use $A_\theta^2 - T_\theta^2 = z(u-k)^2W^2(u^2+2uk)$.

3 The angular difference: nineteen functions

The following calculation clears the intermediate state ratio completely. Define positive source constants

$$A_u = 1/k,\quad A_v = 1/(\sqrt{\sigma}q),\quad A_z = \sqrt{\sigma}/(kq),\quad M_0 = k^2q,\quad a_* = \lambda k,$$

and write $r_* = \sqrt{wW|Z|}$. Eight cleared roots are

$$\begin{aligned}
f_1 &= M_0(A_uA_vA_zuw|Z| - W),\\
f_2 &= \sqrt{2}M_0(A_vA_zw|Z| - A_uuW),\\
f_3 &= M_0\{(\sqrt{a_*} + 1/\sqrt{a_*})A_uA_vuwW - \sqrt{a_*}A_uA_zu|Z| - A_z|Z|/\sqrt{a_*}\},\\
f_4 &= \sqrt{A_u}\sqrt{u}\,f_3,\\
f_5 &= \sqrt{a_*}M_0(W - A_vA_zw|Z|), \qquad f_6 = \sqrt{A_u}\sqrt{u}\,f_5,\\
f_7 &= \sqrt{2a_*A_vA_z}M_0(A_uu - 1)r_*, \qquad f_8 = \sqrt{A_u}\sqrt{u}\,f_7.
\end{aligned}$$

For $Z \geq 0$, $\Delta_A = \sum_{i=1}^{8} f_i^2$. To verify this, put $U = A_uu$, $V = A_vw$, $Z_* = A_z|Z|/W$, $M = M_0W$; the roots are M times the eight roots obtained by expanding every $U+1$ factor in the identity

$$[2 + a_*(U+1)]\Psi + U^2Q_*^2 + [2U + (U+1)/a_*]R_*^2 - T_*^2$$

$$= (T_* + UQ_*)^2 + 2Q_*^2 + (U+1)(\sqrt{a_*}U(V - Z_*) - R_*/\sqrt{a_*})^2$$
$$+ a_*(U+1)(1 - VZ_*)^2 + 2a_*(U+1)VZ_*(U-1)^2,$$

where $\Psi = 1 + U^2V^2 + U^2Z_*^2 + V^2Z_*^2 - 4UVZ_*$, $Q_* = VZ_* - U$, $R_* = Z_* - UV$, and $T_* = U^2 - 1$. It is a polynomial identity by multiplication.

For $Z < 0$, add three squares. Put

$$C_2 = 4, \quad C_1 = 8a_* + 8 + 4/a_*, \quad C_0 = 16 + 8a_* + 4/a_*.$$

Their roots are

$$M_0\sqrt{A_uA_vA_z}\,\mathbf{1}_{Z<0}\big(\sqrt{C_2}A_uu^{3/2}r_*, \quad \sqrt{C_1A_u}\,ur_*, \quad \sqrt{C_0}\sqrt{u}\,r_*\big).$$

Indeed changing Z_* to $-Z_*$ increases the left side by $M^2UV|Z_*|(C_2U^2 + C_1U + C_0)$. This also proves the exact negative-state-sector coefficients. Every displayed source factor is a real closed field square root in the source factor.

The resulting nineteen testing functions, including overlaps with $\mathcal{V}$, are

$$\mathcal{A} = \{W, uw|Z|, w|Z|, uW, uwW, u|Z|, |Z|, u^{3/2}wW, u^{3/2}|Z|, \sqrt{u}|Z|, \sqrt{u}W, \sqrt{u}w|Z|,$$
$$ur_*, r_*, u^{3/2}r_*, \sqrt{u}r_*, \mathbf{1}_{Z<0}u^{3/2}r_*, \mathbf{1}_{Z<0}ur_*, \mathbf{1}_{Z<0}\sqrt{u}r_*\}.$$

Multiply the eleven root formulas above by the pure-source factor $\sqrt{cd_G/S}$ for their contribution to F_w. Every cleared formula is defined at $W = 0$ and $Z = 0$. After substituting squares for $u, w, W, |Z|$, the identities are polynomial; therefore they include those literal boundaries, without a limit or a state-coordinate inverse in the final formulas.

4 The circuit remainder: fourteen functions

Set the positive source constants

$$s_1 = \sqrt{\lambda\sigma}, \quad s_2 = \sqrt{\rho}, \quad s_3 = \sqrt{\lambda}\,xq, \quad s_4 = \sqrt{\lambda x}/q, \qquad g_0 = \sqrt{\lambda x}.$$

Define seven testing functions

$$t_1 = u^{3/4}w^{1/4}W^{1/4}Z_+^{3/4}, \quad t_2 = u^{3/4}w^{3/4}W^{3/4}Z_+^{1/4},$$
$$t_3 = u^{1/4}w^{1/4}W^{3/4}Z_+^{1/4}, \quad t_4 = u^{1/4}w^{3/4}W^{1/4}Z_+^{3/4},$$
$$t_5 = u^{3/4}w^{1/2}W^{1/2}Z_+^{1/2}, \quad t_6 = u^{1/4}w^{1/2}W^{1/2}Z_+^{1/2},$$
$$t_7 = \sqrt{uwWZ_-}.$$

An explicit full-sector SOS is

$$\mathscr{C}_+ = r_1^2 + r_2^2 + r_3^2 + r_4^2,$$
$$r_1 = \sqrt{2g_0}(\sqrt{s_1}t_1 - \sqrt{s_2}t_2), \quad r_2 = \sqrt{2g_0}(\sqrt{s_3}t_3 - \sqrt{s_4}t_4),$$
$$r_3 = 2\sqrt{g_0}\{(s_1s_2)^{1/4}t_5 - (s_3s_4)^{1/4}t_6\}, \qquad r_4 = 2\sqrt{\lambda x}\,t_7.$$

The first three roots vanish for $Z \le 0$ and the last vanishes for $Z \ge 0$. They also vanish wherever $WZ = 0$. On $Z > 0$ they are precisely the separated two-pairing four-monomial identity; on $Z < 0$ the remainder is $4\lambda xuwWZ_-$.

There is no point-dependent scalar reserve at this step:

$$(p_C + a_C(\delta - u))\mathscr{C}_+ = \sum_{j=1}^{4}(\sqrt{p_C}r_j)^2 + \sum_{j=1}^{4}(\sqrt{a_C}\sqrt{\delta - u}\,r_j)^2.$$

Thus the fixed circuit feature list is exactly the seven t_i and their seven products with $\sqrt{\delta - u}$.

5 The evaluated union and machine-readable data

For the main Gram take the seven nonconstant monomials

$$\mathcal{B} = \{w, w^2, \sqrt{u}, u, uw, u^2, u^2w\}, \qquad \gamma_7 = \frac{1}{7\,2^{38}}.$$

The main list is $\mathcal{V}$ times the fifteen fixed factors $1, \sqrt{\gamma_7 + b}, \sqrt{\gamma_7 - b}$ for $b \in \mathcal{B}$, giving 180 functions. Its 144 fixed directions are e_i, $e_i + e_j$, and $e_i - e_j$, with the exact scalar-channel formulas of `fixed_box_c1.tex`, using $7A_*\gamma_7 = c_0$.

Theorem 1. *On the stated independent rectangle, the full real-state parent F_w is a sum of squares of source-linear combinations of one fixed list of* 220 *labelled testing functions: the* 180 *main functions, the seven ordinary functions, the nineteen angular functions, and the fourteen circuit functions. Repeated functions are harmless; the real dimension of their span is at most* 220*. There is an explicit construction using at most* 2196 *square roots:* $144 \cdot 15$ *main roots, seventeen ordinary roots, eleven angular roots, and eight circuit roots. The same list, composed with the fixed gauges above, includes all real state-sign sectors, both testing axes, and the source faces $D = 0$ and $\beta = 0$.*

This theorem follows from the displayed exact decomposition and the separated coefficient formulas. It needs no Gram factorization with mixed radicals. Without using the physical reality reduction, apply the theorem separately to the real and imaginary state amplitudes; their union has at most 440 functions and 4392 roots. The parent is real quadratic in the state, so those two contributions sum to its complex-state value.

The adjacent `c1_retained_union_checks.py` writes all 220 labelled features, all 144 main directions, the main coefficient matrix and root formulas to `c1_positive_lambda_local220.json`. It verifies the two normal identities, the angular identity and signed correction, the circuit identity, the scalar circuit split, the whole-parent assembly, the main matrix support, the fixed-direction identity, and the feature/root counts by exact symbolic arithmetic. This is an evaluated local testing dictionary, not a global full-set affine pencil: the other branches and global cover selection remain outside its scope.